

ARTICLE



Effects of high ambient temperatures on liver function in an occupational population: Examining the mediation role of inflammatory cells and moderation role of age factors

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BACKGROUND: Occupational exposure to high ambient temperature may affect liver function, but pathways remain unclear.

OBJECTIVE: To examine associations between high ambient temperature and liver enzymes among steelworkers, and to evaluate mediation by inflammatory cells and moderation by age.

METHODS: In this cross-sectional study of 1111 steelworkers, high ambient temperature exposure was classified at the workshop level using wet-bulb globe temperature (WBGT) per Chinese standards (GBZ 2.2-2007): time-weighted average WBGT ≥ 25 °C (high-temperature) vs. <25 °C (control). Alanine aminotransferase (ALT), aspartate aminotransferase (AST), and inflammatory cell counts were measured from venous blood. Multivariable generalized linear models estimated associations; mediation (via lymphocytes) used bootstrap confidence intervals; age moderation was tested.

RESULTS: Compared with controls, the high ambient temperature exposure group exhibited elevated levels of ALT and AST, as well as higher counts of white blood cells, lymphocytes, monocytes, and eosinophils ($P < 0.05$). High ambient temperature exposure emerged as independently associated with ALT ($\beta = 0.384$, 95%CI: 0.222, 0.546) and AST ($\beta = 0.238$, 95%CI: 0.065, 0.410) levels. Lymphocyte counts statistically mediated the relationship between high ambient temperature exposure and elevated serum ALT ($\beta = 0.062$, 95%CI: 0.026, 0.104) and AST ($\beta = 0.046$, 95%CI: 0.009, 0.092) levels. Additionally, age only moderated the path between high ambient temperature exposure and lymphocyte count ($\beta = -0.192$, 95%CI: -0.316, -0.067).

SIGNIFICANCE: Occupational heat was associated with higher ALT/AST; lymphocyte-related indirect association was observed and attenuated with age. Longitudinal and mechanistic studies are needed to assess causality.

IMPACT: Findings support WBGT-based heat monitoring, routine ALT and AST screening in hot posts, and targeted acclimatization hydration and work-rest protocols, especially for younger workers, while future studies test whether enhanced heat controls improve liver outcomes.

Keywords: High ambient temperature exposure; Liver function; Inflammatory cells; Mediating effects; Moderating effects

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INTRODUCTION

Globally, a substantial proportion of the workforce is exposed to occupational heat stress, a major public health concern that is projected to worsen due to climate change. This issue is particularly pressing in low- and middle-income countries, where workers are often engaged in labor-intensive activities under extreme heat conditions. The adverse effects of occupational heat stress are well-documented, encompassing a range of health and productivity outcomes [1]. The International Labour Organization estimates that over 2.4 billion workers are exposed to excessive heat annually [2]. Exposure to high ambient temperatures is known to have detrimental effects on human health, contributing

to approximately 0.64% of annual deaths in China [3, 4]. It's worth noting that China boasts the world's largest working population. In 2021, the National Bureau of Statistics reported 74,652 million occupational workers in China, accounting for 52.85% of the total population. Occupational health concerns related to heat have long persisted, especially in professions involving strenuous physical labor or exposure to high ambient temperatures [5, 6]. Research evidence has indicated that long-term exposure to high ambient temperatures can result in adaptive damage to various bodily systems, including the circulatory system [7], nervous system [8], and digestive system [9], and in severe cases, it may lead to heatstroke and other related illnesses [10, 11]. While the

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impacts of heat stress on cardiovascular and renal outcomes have been relatively well-documented, research focusing on its effects on liver function within occupational populations remains limited, especially in occupational population.

The liver is central to detoxification, energy metabolism, and regulation of immune-inflammatory responses [12]. These roles become critically relevant in the context of occupational heat stress, as the liver is central to processing metabolic waste generated by increased physiological strain, managing the systemic inflammatory cascade triggered by heat exposure, and handling potential co-exposures to hepatotoxicants in industrial settings. In clinical practice, serum alanine aminotransferase (ALT) and aspartate aminotransferase (AST) are sensitive and widely used biomarkers for the early detection of hepatocellular injury. A study based on the 2019 Global Burden of Disease (GBD) database shows that liver diseases claim the lives of over 2 million people globally each year, with nearly 300 million individuals affected by liver disease in China alone [13]. Among the many risk factors for impaired liver function, the living environment, including unhealthy lifestyles [14], environmental pollution [15], and occupational exposure [16, 17] has received increasing attention for its potential to induce abnormal liver function. Liver injury frequently emerges as a common complication of heatstroke (HS) stemming from exposure to high ambient temperatures and is often the direct cause of death in affected patients [18]. However, much less is known about the potential subclinical effects of chronic, occupational heat exposure at levels below those required to cause heatstroke. Animal experimental studies have shown that long-term exposure to high temperatures can cause damage to liver tissue: Gupta et al. used Sprague Dawley (SD) rats as a model to expose them to moderate but sustained heat stress and found that compared with control rats, The liver function test levels of rats in the heat exposure experimental group decreased, the levels of free radicals and inflammatory factors increased, the structural deterioration of liver lobules increased, and the mitochondrial structure was destroyed [19]. Most of the research on the impact of high temperature on the liver in the population focuses on the evaluation of the impact on the liver when high-temperature work leads to more serious outcomes. Therefore, it is imperative to investigate the effects of chronic high ambient temperature exposure on these key liver function indices among occupational workers. Such investigations contribute significantly to our understanding of heat-induced liver damage and the development of targeted protective measures.

This study aimed to investigate the correlation between high ambient temperatures and liver function indicators, specifically ALT and AST levels, among workers at a central China steel plant. Additionally, by analyzing the mediating and moderating effects of high ambient temperature exposure on liver function, this research provides valuable insights and directions for further investigation into the mechanisms. These findings offer a basis and theoretical framework for the effective prevention and management of heat-induced liver effects.

MATERIALS AND METHODS

Study subjects

A total of 1111 participants were enrolled in this study. The high ambient temperature exposure group comprised 606 participants, while the control group, consisting of 505 individuals, worked in the same factory without exposure to high ambient temperatures. Inclusion required age 18–60 years and ≥ 1 year of employment at the plant. Exclusion criteria were documented organic diseases of major organs (e.g., liver or kidney disease) or malignancy; pregnancy or lactation; and incomplete questionnaires/records or missing key exposure or outcome data.

The research protocol received ethical approval from Zhengzhou University's ethics committee (ZZUIRB2021-153). Before participating, all participants provided written informed consent.

High ambient temperature exposure assessment

High ambient temperatures were assessed using a comprehensive wet-bulb globe temperature (WBGT) meter in accordance with the Occupational Health Standard of China GBZ/T 189.7—2007. Measurements included dry-bulb, natural wet-bulb, and globe temperatures (via WBGT meter, dry-bulb thermometer, natural wet-bulb thermometer, and black-globe thermometer), from which the WBGT index was calculated per the standard. For workplaces without productive heat sources, three fixed points were measured and averaged; for workplaces with heat sources, 3–5 points were averaged. Where areas differed in thermal or ventilation conditions, two points per area were measured and averaged. For year-round hot work, monitoring was conducted in the hottest summer month; for intermittent hot work, in the hottest month of the project period; and for outdoor work, on sunny days with solar radiation during the hottest month. When the heat source was stable, measurements were taken three times per day (≈ 0.5 h after start, mid-shift, and ≈ 0.5 h before end) and averaged. When the heat source was unstable or cycles were long, a time-weighted average (TWA) WBGT was calculated. Following GBZ 2.2—2007, hot work is classified into four levels by workplace WBGT and exposure duration. Consistent with the GBZ 2.2—2007 definition, workshops with $\text{WBGT} \geq 25^\circ\text{C}$ were designated the high-temperature group, and those with $\text{WBGT} < 25^\circ\text{C}$ the control group.

Liver function measures

Participants were required to fast for a minimum of 8 h and abstain from alcohol consumption for 24 h before providing approximately 5 ml of peripheral blood. The blood was collected using EDTA anticoagulant tubes and tubes containing coagulant. An automatic biochemical analyzer was used to detect inflammatory cell counts, such as white blood cells, lymphocytes, neutrophils, monocytes, eosinophils, and basophils, and liver function indicators, specifically ALT and AST.

Variable selection

The questionnaire survey collected basic information, occupational history (including occupational experience, occupational time, work unit, work workshop, type of work, exposure factors, etc.), past medical history, acute and chronic occupational disease history, family history, medication history, etc, as well as height and weight. The final multivariable models adjusted for sex, employment duration (years), smoking status, alcohol use, and co-exposures measured in the workplace: noise (dBA), total dust (mg/m^3), and carbon monoxide (mg/m^3). According to WHO guidelines, current smokers are defined as individuals who smoke daily for more than a year, and current drinkers are defined as people who drink at least twice a week for at least six months [20].

Statistical analysis

The physical examination data were entered in a double-blind manner using EpiData 3.0 and analyzed using SPSS 25.0. The median and interquartile range were used to describe measurement data that did not follow a normal distribution, while the count value and count percentage (n (%)) were used for the statistical description of the count data. Differences in liver function indicator levels were assessed using the Mann-Whitney U test, while the χ^2 test was used to compare the differences in rates between groups. Multivariate analysis of liver function indicator levels was conducted using a generalized linear model. Mediation analyses were performed using the SPSS program (PROCESS 3.5 v) [21]. The significance of the mediating effect was determined using the bias-corrected percentile Bootstrap method (with 5000 replicate samples) and 95% confidence intervals (95% CI). The two-sided test level was set at $\alpha = 0.05$.

RESULTS

Basic information of subjects

Approximately 87.79% of participants in the exposure group and 69.1% in the control groups were male, with a comparable age distribution in both groups, as more than half of the personnel were younger than 44 years old. Additionally, participants in the exposure group reported a higher prevalence of smoking, alcohol consumption, noise exposure, dust, and carbon monoxide exposure compared to the control group ($P < 0.001$). The basic characteristics of the study participants are presented in Table 1.

Table 1. The demographic characteristics of participants.

Characters	Control group (n = 505)	Exposure group (n = 606)	χ^2 / Z	P
Gender (%)				
Male	350 (69.31)	532 (87.79)	57.500 ^a	<0.001
Female	155 (30.69)	74 (12.21)		
Age (%)				
≤44	296 (58.61)	355 (58.58)	<0.001 ^a	0.991
>44	209 (41.39)	251 (41.42)		
Working duration (%)				
≤24	271 (53.66)	334 (55.12)	0.234 ^a	0.628
>24	234 (46.34)	272 (44.88)		
Blood pressure (%)				
Normal	282 (55.84)	303 (50.00)	3.770 ^a	0.052
Abnormal	223 (44.16)	303 (50.00)		
Smoking (%)				
No	285 (63.47)	256 (49.14)	20.009 ^a	<0.001
Yes	164 (36.53)	265 (50.86)		
Drinking (%)				
No	253 (56.35)	205 (39.35)	27.965 ^a	<0.001
Yes	196 (43.65)	316 (60.65)		
Noise exposure (%)				
No	393 (77.82)	102 (16.83)	414.773 ^a	<0.001
Yes	112 (22.18)	504 (83.17)		
Dust exposure (%)				
No	370 (73.27)	128 (21.12)	302.843 ^a	<0.001
Yes	135 (26.73)	478 (78.88)		
CO exposure (%)				
No	439 (86.93)	243 (40.10)	254.868 ^a	<0.001
Yes	66 (13.07)	363 (59.90)		

Note: a: categorical variables were compared between two groups using the Chi-square test; b: non-normal data were compared between two groups using the Mann-Whitney U test.

Differences in Inflammatory and liver function indicators under high ambient temperature exposure

Inflammatory indicators, including white blood cell (WBC) count, lymphocyte count, monocyte count, and eosinophil count, were significantly higher in the exposure group compared to the control group. Liver function indicators, ALT, and AST also exhibited notable differences. The median ALT level in the exposure group was 26.00 U/L, significantly higher than the control group (20.00 U/L, $P < 0.001$). Similarly, the median AST level in the high ambient temperature exposure group was elevated compared to the control group (24.00 U/L vs. 22.00 U/L) (Table S2).

Distribution of liver function indicators in exposed and control groups across different demographic characteristics

Analysis revealed that the median ALT level in the exposure group (27.00 U/L) was higher than that in the control group (24.00 U/L) ($P = 0.001$), and this difference was observed only in males. Both age groups within the exposure group had significantly higher ALT levels compared to the control group (all $P < 0.005$). Within the smoking group, ALT levels in the exposure group were higher than in the control group ($P < 0.001$ and $P = 0.008$, respectively). For the drinking group, ALT levels in the exposure group were higher than those in the control group (Fig. 1, Table S3).

In contrast to ALT levels, the median AST in the exposure group (23.00 U/L) was higher than that in the control group (21.00 U/L, $P < 0.001$). Within the smoking group, median AST levels for the non-smoking group were 23 U/L, higher than the control group (21 U/L, $P < 0.001$). In the drinking group, median AST levels for the non-drinking group were 22 U/L, also higher than the control group (20 U/L, $P < 0.001$) (Fig. 1, Table S3).

Association of high ambient temperature exposure with liver function indices

Table 2 presents the associations between high ambient temperature exposure and liver function indices in a generalized linear model. Model 1 (crude model) indicated a positive association between high ambient temperature exposure and both ALT ($\beta = 0.194$, $P = 0.001$) and AST ($\beta = 0.174$, $P = 0.006$). After adjusting for covariates such as noise exposure, total dust, CO exposure, sex, age, length of service, blood pressure, smoking, and drinking, high ambient temperature exposure maintained a significant positive association with both ALT ($\beta = 0.384$, $P < 0.001$) and AST ($\beta = 0.238$, $P = 0.008$) (Model 5).

Mediating effect of high ambient temperature on liver function

Spearman's correlation coefficient was used to analyze correlations among high ambient temperature exposure, liver function indices (ALT and AST), and inflammatory cells. Results indicated significant correlations between high ambient temperature exposure and liver function indices, Lymphocyte count, and Monocyte count ($P < 0.001$). Similarly, liver function index (ALT and AST) exhibited significant correlations with Lymphocyte count, Monocyte count, and Basophil count ($P < 0.001$) (Fig. 2 and Table S3).

Mediation analysis results demonstrated that lymphocyte count significantly mediated the association between high ambient temperature exposure and ALT, as well as AST levels (Table S4). Fig. 3 illustrates the mediating effect of high ambient temperature exposure on ALT levels, with lymphocyte count mediating the association ($\beta = 0.062$), explaining 10.86% of the mediating effect. Fig. 4 displays the mediating effect on AST levels, with lymphocyte count accounting for a mediating effect of 0.046 or 12.23%.

Moderating effect of age on liver function during high ambient temperature exposure

Table 3 outlines the moderating effect of age on the relationship between high ambient temperature exposure and liver function index levels. Age was found to moderate the effect on ALT levels during high ambient temperature exposure, with a moderating effect of -0.137, $P = 0.023$. In other words, the effect of high ambient temperature exposure on ALT levels decreased as age increased.

Moderated mediation effects analysis of age on the pathways in the process of high ambient temperature exposure affecting ALT

To further investigate the specific moderated path of age, a moderated mediation effects model was constructed, with the pattern diagram shown in Fig. 5. The analysis revealed that, in the mediating model of high ambient temperature exposure affecting ALT levels, age only moderated the path between high ambient temperature exposure and lymphocyte count ($\beta = -0.192$, 95%CI: -0.316, -0.067) (Table 4). Using a simple slope to illustrate the effect pattern of the moderating variable in the mediating model of the regulation (Fig. 6), the results showed a stronger association between hyperthermia exposure and lymphocyte counts in younger participants compared to their older counterparts.

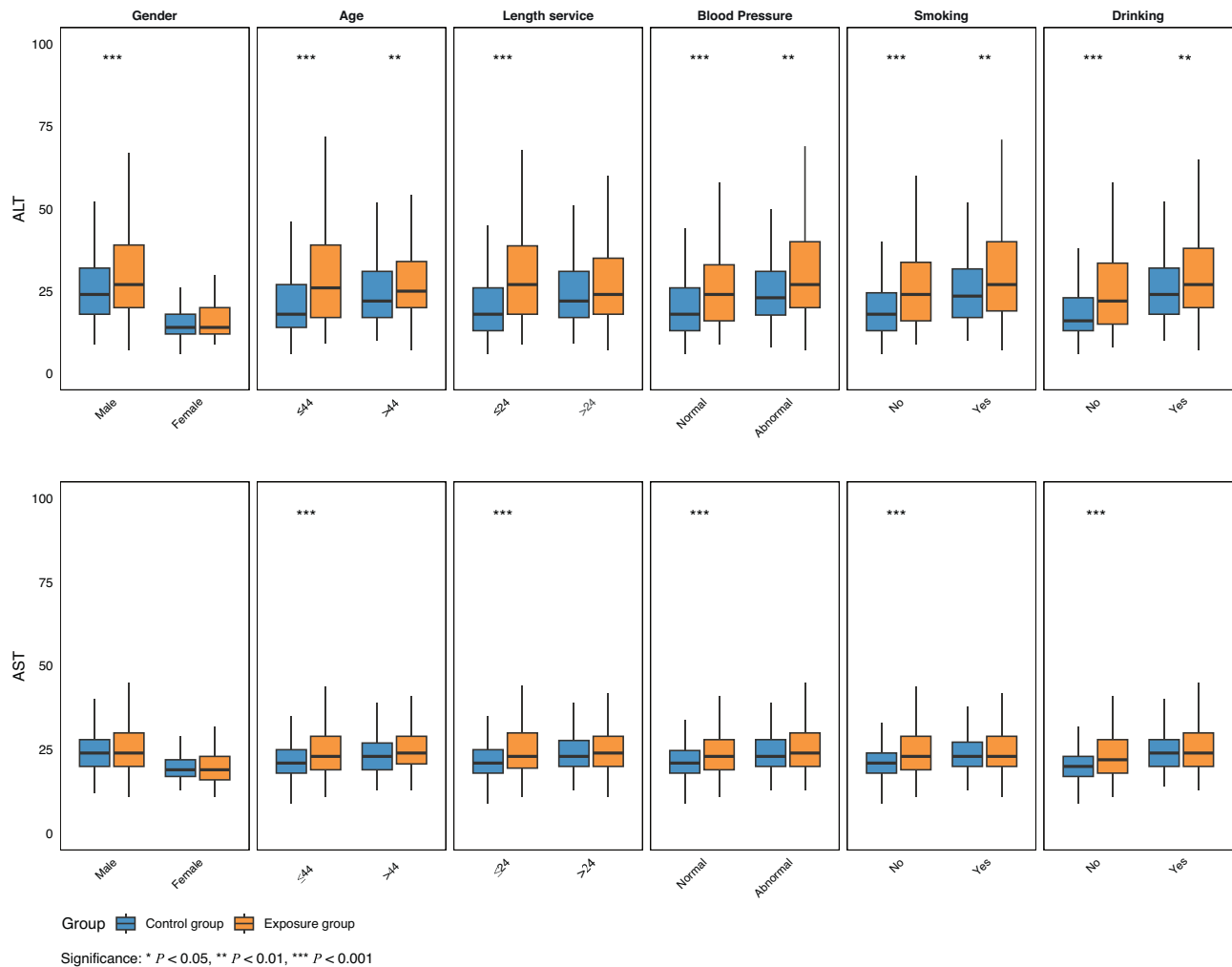


Fig. 1 Serum ALT and AST levels in the exposure group and control group across demographic and lifestyle subgroups. Boxplots show the distributions of ALT and AST in the control group and exposure group, stratified by gender, age, length of service, blood pressure, smoking status, and drinking status. The blue boxes represent the control group, and the orange boxes represent the exposure group.

DISCUSSION

In this cross-sectional study of steelworkers, high ambient temperature which classified using weighted wet-bulb globe temperature (WBGT) at the workshop was associated with higher serum ALT and AST and with higher white blood cell, lymphocyte, monocyte, and eosinophil counts. Furthermore, lymphocyte counts were identified as mediators in this association, while age acted as a modifier of the association between high ambient temperature exposure and lymphocyte counts. Our results are consistent with occupational evidence that heat exposure relates to hepatic biomarkers where liver injury is frequently observed. In a study on firefighters, exposure to fire emissions and heat was associated with significantly altered values of biomarkers related to liver injury, indicating the potential impact of their occupational exposure on liver health [22]. Solar ultraviolet radiation and heat stress are major concerns for outdoor workers. Heat stroke, the most severe form of heat stress, can cause irreversible damage to multiple organs including the liver [23]. Our findings align with prior research, demonstrating a positive association between high ambient temperature exposure and ALT and AST levels. Importantly, our findings address subclinical, chronic occupational heat below HS thresholds in an industrial setting, complementing the stronger evidence base for liver damage outcomes. Recent studies have delved into the mechanisms of liver damage caused by high ambient temperature exposure. Heat stress triggers a cascade of

physiological and metabolic reactions in response to extreme temperatures, such as those experienced during high ambient temperature exposure [24]. The liver is a primary target organ affected by heat stress damage, resulting in protein inactivation and structural damage in liver cells [25]. These changes are reflected in early increases in serum total bilirubin, alanine aminotransferase, aspartate aminotransferase, and other signaling factors [26]. Prolonged heat stress can cause or aggravate cell damage or dysfunction [27]. The liver, rich in blood flow, undergoes redistribution of blood during heat stress, leading to decreased blood flow through the liver [28]. This aggravates liver cell hypoxia, activates proteases, and triggers liver macrophages. This cascade of events damages mitochondrial structures reduces superoxide dismutase synthesis, and increases the production and accumulation of oxygen free radicals, resulting in aggravated oxidative damage to the liver [29, 30].

Lymphocyte count emerged as a mediator in the relationship between high ambient temperature exposure and liver function indexes, specifically ALT and AST. This finding suggests that high ambient temperature exposure may lead to changes in lymphocyte counts, subsequently influencing ALT and AST levels. Changes in lymphocyte counts are often associated with chronic inflammatory reactions or viral infections [31], consistent with the hypothesis that steel mill workers, exposed to long-term high-temperature environments, experience chronic liver inflammation.

Table 2. Multivariate analysis of the effect of high ambient temperature on liver function index in different models.

Models	Liver function index	β (95%CI)	P
Model1 ^a			
	ALT	0.194 (0.076, 0.312)	0.001
	AST	0.174 (0.049, 0.300)	0.006
Model2 ^b			
	ALT	0.328 (0.179, 0.476)	<0.001
	AST	0.224 (0.066, 0.381)	0.005
Model3 ^c			
	ALT	0.251 (0.111, 0.390)	<0.001
	AST	0.201 (0.053, 0.348)	0.008
Model4 ^d			
	ALT	0.260 (0.125, 0.395)	<0.001
	AST	0.187 (0.045, 0.330)	0.010
Model5 ^e			
	ALT	0.384 (0.222, 0.546)	<0.001
	AST	0.238 (0.065, 0.410)	0.007

Note: β is the standard regression coefficient of high ambient temperature exposure factors obtained after standardization; a: Model 1 adjusted the variables of sex, age, length of service, blood pressure, smoking and drinking; b: Model 2 adjusted noise, sex, age, length of service, blood pressure, smoking, drinking and other variables; c: Model 3 adjusted the variables of total dust, sex, age, length of service, blood pressure, smoking and drinking; d: Model 4 adjusted CO exposure, sex, age, length of service, blood pressure, smoking, drinking and other variables; e: Model 5 adjusted noise exposure, total dust, CO exposure, sex, age, length of service, blood pressure, smoking, drinking and other variables.

In contrast, other inflammatory cells like neutrophils, are typically associated with acute inflammation [32, 33], while eosinophils are more common in parasitic [34] or allergic diseases such as asthma [35]. These cells did not significantly play a mediating role in the association between high ambient temperature exposure and liver function indicators, which was consistent with our findings. As previously mentioned, the human body experiences heat stress when exposed to high ambient temperature exposure, leading to a series of adaptive [36], stress, and immune-inflammatory responses as it compensates for heat-induced stress [37]. Notably, the inflammatory response constitutes a vital component of this compensatory process. This inflammatory response activates lymphocytes, leading to an increase in their numbers within the body [38]. We emphasize that these mechanistic claims derive from prior literature rather than direct measurements in our dataset; in our study, the statistical indirect association via lymphocytes is consistent with, but does not prove, an immune-mediated pathway.

Finally, our mediation analyses demonstrated that age primarily regulates the pathway linking high ambient temperature exposure to increased lymphocyte counts, such that the indirect effect on ALT/AST via lymphocytes is stronger in younger workers and attenuates with age. Given the cross-sectional design, mediation reflects a statistical decomposition (total = direct + indirect association) and does not establish temporality or causality. The age interaction implies heterogeneity of the association: younger workers showed stronger exposure-related lymphocyte differences. In light of the findings discussed above, this study suggests that older individuals in the exposed group may exhibit improved adaptation or acclimatization to high ambient temperatures, thereby enhancing their ability to manage the inflammatory response induced by such conditions, ultimately reducing liver damage. This pattern suggests two, non-mutually exclusive explanations. First, heat acclimatization may develop with

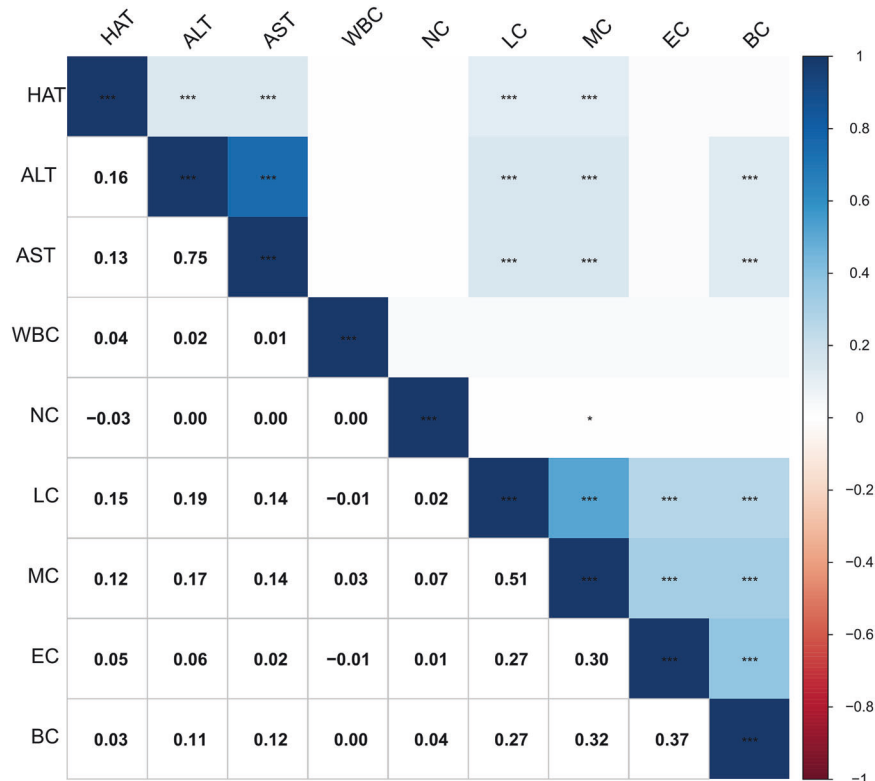


Fig. 2 Correlation among high ambient temperature exposure, liver function index and inflammatory cells. HAT high ambient temperatures, ALT Alanine aminotransferase, AST Aspartate aminotransferase, WBC White blood cell count, NC Neutrophil count, LC Lymphocyte count, MC Monocyte count, EC Eosinophil count, BC Basophil count.

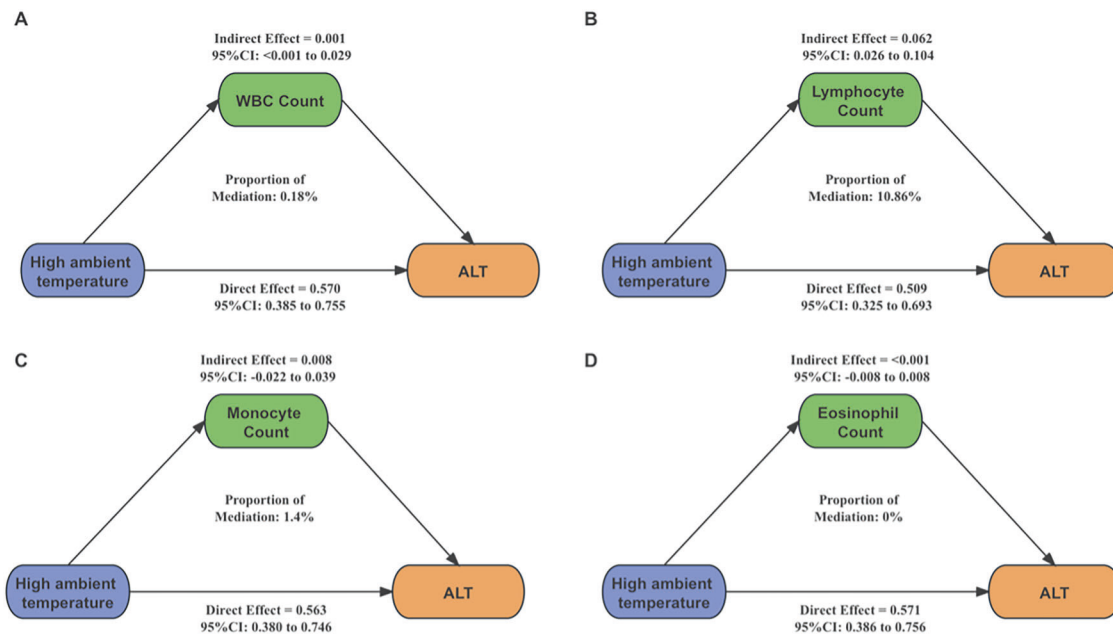


Fig. 3 Mediating effect model of high ambient temperature exposure on ALT levels. ALT alanine aminotransferase, AST aspartate aminotransferase, SE standard error, a adjusted for noise exposure, total dust exposure, CO exposure, gender, age, length of service, blood pressure, smoking, drinking and other confounding factors.

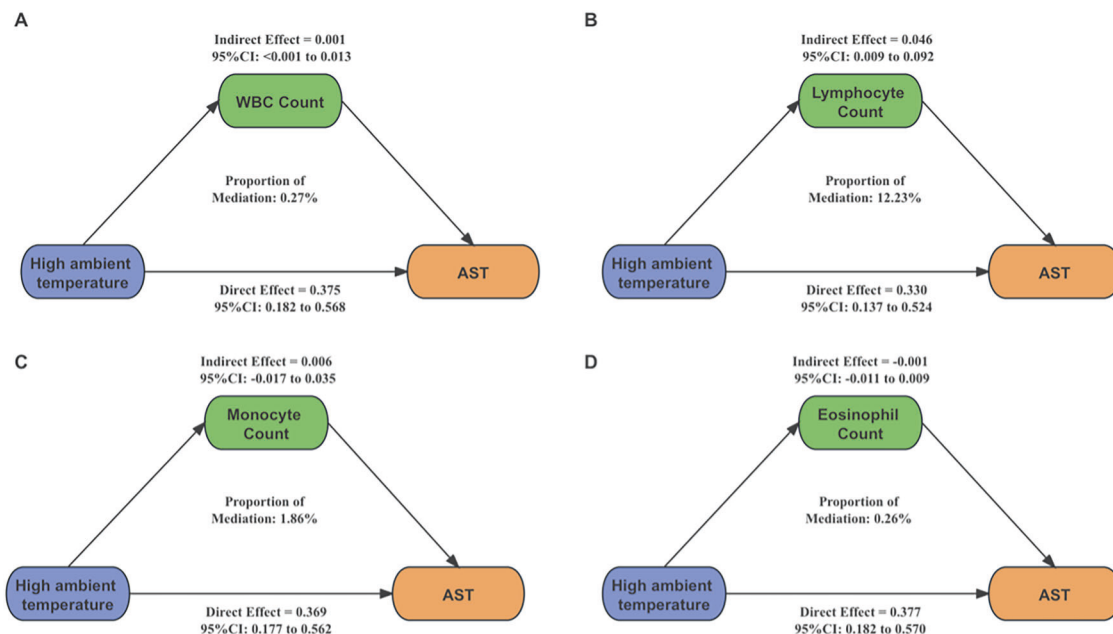


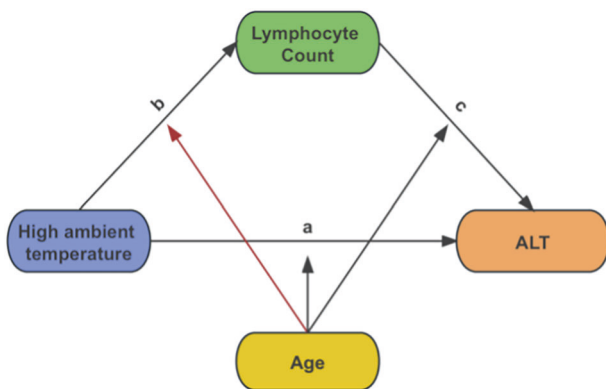
Fig. 4 Mediating effect model of high ambient temperature exposure on AST levels. Note: ALT alanine aminotransferase, AST aspartate aminotransferase, SE standard error, a adjusted for noise exposure, total dust exposure, CO exposure, gender, age, length of service, blood pressure, smoking, drinking and other confounding factors.

prolonged exposure, improving thermoregulatory efficiency and enabling more adaptive behaviors, thereby dampening heat-related inflammatory responses and their downstream impact on liver enzymes among older workers. In light of the findings discussed above, this study suggests that older individuals in the exposed group may exhibit improved adaptation or acclimatization to high ambient temperatures, thereby enhancing their ability to manage the inflammatory response induced by such conditions, ultimately reducing liver damage. Second, a healthy worker survivor effect may operate: older workers who remain in heat-

intensive jobs could represent a selected, healthier subset, while more susceptible individuals may have exited or been reassigned earlier, which would attenuate observed associations with age in cross-sectional data. It's important to note that the research subjects in the high ambient temperature exposure group were relatively concentrated in terms of age, with a median and interquartile range M (P25, P75) is 44.00 (40.00, 47.00) respectively. Workers in this age group generally exhibit better physical fitness and a higher level of social benefit output, which differs from the elderly population in general. As a result, the aging factors have a

Table 3. Effect of age on liver function during high ambient temperature exposure.

Outcome Variable	β	β (95%CI)	
		Lower	Upper
ALT			
X: High ambient temperature	0.565	0.379	0.750
W: Age	0.002	-0.093	0.097
X * W	-0.137	-0.256	-0.019
F	5.169		
R ²	0.005		
P	0.023		
AST			
X: High ambient temperature	0.382	0.187	0.577
W: Age	0.031	-0.069	0.130
X * W	-0.054	-0.179	0.070
F	0.730		
R ²	0.001		
P	0.393		

**Fig. 5** Moderated mediation model of the association between high ambient temperature and ALT levels. Path a represents the association between high ambient temperature and ALT levels, path b represents the association between high ambient temperature and lymphocyte count, and path c represents the association between lymphocyte count and ALT levels. Age was included in the model as a moderator, and the red arrow indicates its moderating effect on the association between high ambient temperature and lymphocyte count.

lesser impact on their immune system. Another crucial factor that plays a regulatory role is high ambient temperature exposure itself, which represents a physical occupational hazard, which is different from exposure to general chemical occupational hazards that may accumulate within the body. High ambient temperature exposure does not lead to such accumulation effects. Instead, those exposed to high ambient temperatures over extended periods tend to acquire heat acclimatization, significantly improving their body's tolerance to these conditions and, consequently, reducing liver damage [39]. Our results shows that high ambient temperatures lead to higher ALT/AST, partly via lymphocyte counts and with stronger effects in younger workers—have practical implications for occupational health programs in hot settings. Prioritize younger and newly hired workers and those returning after prolonged absence, given the stronger indirect effects observed in these groups. Incorporate liver function tests (ALT/AST) and complete blood counts (lymphocytes) alongside existing heat-stress metrics. Consider post-incident testing after heat-related events or symptom reports ; Provide shade, ventilation, and spot-cooling at task hot-spots; reduce radiant heat where feasible; ensure water/electrolyte stations are proximal to work areas. Targeted training and acclimatization: Emphasize acclimatization plans for new workers, and reinforce behaviors that reduce strain.

In summary, this study focused on steel mill workers as its research subjects to investigate the correlation between high ambient temperature exposure and the levels of liver function indicators, specifically ALT and AST. Additionally, we explored the mediating role of inflammatory cells and the moderating role of age in the context of high ambient temperature exposure affecting ALT and AST levels. While previous research primarily relied on animal experiments to assess the effects of high ambient temperature exposure on liver function, our study concentrated on occupational workers exposed to these conditions, rendering the conclusions more applicable at the population level. Nonetheless, this study has certain limitations. As discussed, cross-sectional design precludes causal inference between occupational high ambient temperature exposure and serum liver enzyme levels. Second, despite multivariable adjustment, residual confounding could not be excluded; important covariates such as hydration status, BMI, medication use (including potentially hepatotoxic drugs), and other factors affecting the liver were not obtained. Third, ALT/AST and inflammatory cell counts were measured once, which may not capture short-term biological variability. Fourth, high ambient temperature is not a continuous exposure and varies by task, shift, and day, which may bias the association toward or away from the null hypothesis. In addition, our exposure definition followed GBZ thresholds to align with local regulation and occupational practice, increasing the policy relevance of our findings for Chinese workplaces. We acknowledge that reporting continuous WBGT would allow direct mapping to widely used international schemes (ISO 7243/ISO

Table 4. Effects of age on the pathways in the process of high ambient temperature affecting ALT.

Outcome Variable	β	SE	t / Z	P	β (95%CI)	
					Lower limit	Upper limit
Lymphocyte Count						
High ambient temperature * Age	-0.192	0.063	-3.027	0.003	-0.316	-0.067
ALT						
Lymphocyte Count * Age	0.017	0.029	0.577	0.564	-0.040	0.070
High ambient temperature * Age	-0.117	0.061	-1.908	0.057	-0.237	0.003

Note: SE: standard error.

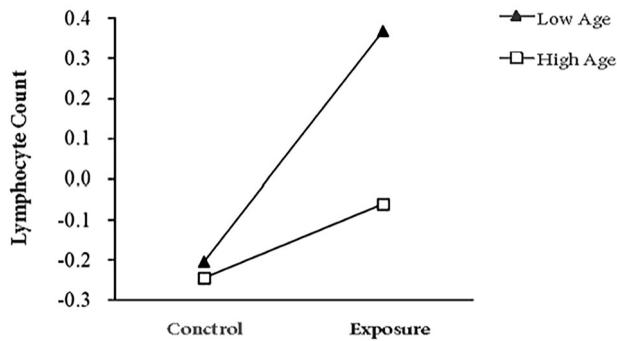


Fig. 6 Interaction of age and high ambient temperature exposure on lymphocyte count. Simple slope plot showing the moderating effect of age on the association between high ambient temperature exposure and lymphocyte count. Triangles indicate the low-age group, and squares indicate the high-age group.

7933, ACGIH TLVs, NIOSH guidance). Future research should incorporate longitudinal designs, repeated biomarker assessments, and personal heat monitoring (e.g., continuous WBGT), alongside systematic collection of hydration, BMI, medication use, and liver history; methods that address time-varying selection (e.g., inverse probability weighting) and cross-walks to international frameworks will strengthen causal interpretation and external validity.

In conclusion, our cross-sectional study of steelworkers suggests that occupational exposure to high ambient temperatures is associated with higher serum ALT and AST and seems to act, at least in part, through increased lymphocyte counts; this effect appears stronger in younger workers and attenuates with age. Based on these findings, routine liver function screening in hot jobs, targeted heat-health promotion for younger workers, and the integration of liver risk indicators into occupational heat-health guidelines seem warranted.

DATA AVAILABILITY

Data will be made available upon request.

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AUTHOR CONTRIBUTIONS

Wei Wang and Mengqing Yan conceived of the study. Mengqing Yan and Bin Yang designed the study. Mengqing Yan wrote the main manuscript text. Bin Yang, Menghan Li, Wenjun Kang and Yongli Yang collected the data and did the data analysis. All authors read and revised the manuscript and gave final approval for publication.

COMPETING INTERESTS

The authors declare no competing interests.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

The study was approved by ethical approval from Zhengzhou University's ethics committee (ZZUIRB2021-153). Informed consent was obtained from all participants prior to study participation. This study was conducted in accordance with the principles of the Declaration of Helsinki.

ADDITIONAL INFORMATION

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